

Visual Place Recognition

IE Tower

Project Overview & Recording Guide for the Team

Computer Vision | Spring 2026 | IE University

1. What We're Building

We are building an **image search engine for place recognition** inside the IE Tower. Given a query photo taken anywhere in the building, the system retrieves the most similar images from a database and tells you **where in the Tower that photo was taken**.

Think of it like a visual GPS that works indoors: you snap a photo, the system matches it against its database, and returns the location (floor + specific area).

How It Works (Simple Version)

1. **Record video** of every location in the IE Tower (that's what this guide is for).
2. **Extract frames** from those videos (1 frame per second).
3. **Generate embeddings** — each frame gets converted into a numerical vector (a list of 2048 numbers) using a pre-trained neural network (ResNet50). Similar-looking places produce similar vectors.
4. **Build an index** — all vectors are stored in a FAISS index for fast search.
5. **Query** — when someone uploads a new photo, we extract its vector and find the closest matches in the index.

Key idea: the quality of the final system depends directly on the quality and consistency of the recorded dataset. That's why we need everyone following the same recording protocol.

2. What We Need to Record

The dataset covers **indoor spaces from floor -4 to floor 22** of the IE Tower. We need a balanced mix of:

- **Distinctive spaces:** cafeterias (floors 4, 10, 16, 22), reception (floor 0), robotics lab, athletic center, auditorium/theater, library.
- **Academic spaces:** classrooms of different types (curved desks, flat, large lecture halls), study rooms, meeting rooms.
- **Circulation areas:** selected hallways with visual landmarks, elevator lobbies, staircases.
- **Basement areas:** labs, corridors, piano room, distinctive bathrooms (floors -4, -3, -2 only).

Each location must be recorded in **at least 2 lighting conditions**: day/afternoon and night. This teaches the model that the same place can look different depending on when the photo was taken.

About **~10% of footage** should come from a laptop webcam (not Blackmagic). This makes the system more robust to real-world queries where someone might use a worse camera.

3. Recording with Blackmagic Camera

We all use the same app to keep recordings consistent. Download it free from the App Store:

Blackmagic Camera — search "Blackmagic Camera" on the App Store (iOS).

Required Settings

Open the app and configure **every setting manually** before recording. Do NOT leave anything on auto.

Setting	Value	Why
Resolution	720p	Enough quality, small file size
Frame Rate	30 FPS	Standard; we extract 1 fps anyway
Codec	HEVC (H.265)	Smaller files, same quality
Lens (OBJ)	24mm	Wide enough for indoor spaces
Shutter (OBT)	1/60	Lock it — no motion blur variance
ISO	Manual	Adjust per location; keep as low as possible (100-400)
White Balance	~4300K	Lock it — indoor lighting consistency
Tint	0	Neutral
Focus	Manual	Lock before recording each clip
HDR	OFF	We don't want HDR processing
Orientation	Horizontal only	Landscape — never vertical

Important: The white balance in the Blackmagic app screenshot shows 5980K — that's too warm for most indoor IE Tower spaces. Set it closer to **4300K** for consistent indoor fluorescent/LED lighting. Lock it (tap the lock icon).

4. How to Record

Before You Start

- Open Blackmagic Camera and verify ALL settings match the table above.
- Hold the phone **horizontally** (landscape mode).
- Make sure the area is representative — don't record during unusual events or with obstructions.
- Lock focus on the main subject/area before pressing record.

During Recording

- **Walk slowly and steadily.** No sudden movements. Think of it like a smooth dolly shot.
- **Record 15-30 seconds per clip** — one clip = one specific location.
- **Cover the space:** pan gently across the area so the model sees different viewpoints of the same place.
- **Avoid filming people's faces** when possible — they add noise to the embeddings.
- **Don't tilt** the camera up/down dramatically. Keep it roughly eye-level, horizontal.

- **Include visual landmarks:** room numbers on doors, signs, distinctive furniture, windows, structural elements.

Lighting Conditions

Each location needs at least 2 recordings under different lighting:

Condition	When	What Changes
Day / Afternoon	Natural daylight hours	Windows bright, mixed light
Night	After dark / blinds closed	Only artificial light, warmer tones

5. Laptop Webcam Footage (~10%)

For about 10% of locations, also record a short clip using your laptop's built-in webcam. This doesn't need to follow the Blackmagic settings — that's the point. We want the model to handle lower-quality, different-looking input.

Just open your laptop camera (Photo Booth, OBS, or any recorder), point at the location, and record 10-15 seconds. Save as .mov or .mp4.

6. After Recording — What Happens Next

Once you finish recording, the workflow is:

1. **Split videos into clips** — each clip = one location, one lighting condition.
2. **Label each clip** — using our labeling convention (see Document 2).
3. **Extract frames** — our script pulls 1 frame/sec from each clip automatically.
4. **Upload to the repo** — push your frames and metadata to the shared GitHub repo.

The labeling convention and detailed workflow for splitting/uploading are in **Document 2: Location Labels & Workflow**.

7. Quick Reference Card

Item	Value
App	Blackmagic Camera (iOS, free)
Resolution	720p
FPS	30
Codec	HEVC
Lens	24mm
Shutter	1/60 (locked)

Item	Value
ISO	Manual, as low as possible
White Balance	~4300K (locked)
Focus	Manual (lock before each clip)
Orientation	Horizontal ONLY
Clip Length	15-30 sec per location
Lighting	2 conditions minimum per location
Webcam	~10% of locations, any settings

Questions? Ping the group chat. Consistency is everything for this project.